

Water Quality Monitoring of Peruvian Amazon Based in the Internet of Things

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Abstract—This paper presents a water quality monitoring system using low power hardware based in the Internet of Things. Six sensor nodes for water quality monitoring in pisciculture are installed in three region of the Peruvian amazon. The sensor node is a modular hardware composed by control, communication and sensor module that measure parameters such as pH, conductivity and temperature. All data information is managed by software platform designed using open source tools, data information allows to control water pollution and their effect in the pisciculture. Data acquired is used by the researchers of the Instituto de Investigaciones de la Amazonía Peruana to study the health and growing population of paiche fish.

Keywords—Sensors, Sensor Node, ZigBee, GPRS, WSN, IoT.

I. INTRODUCTION

Monitoring water quality is critical for human health and the health of the natural ecosystem. Different problematics as the environmental degradation and climatic change have put in risk the water quality of rivers, streams and lakes, these water source are known surface water which is often used in different activities; human consumption, irrigation, farming, animal husbandry and pisciculture. The principal pollutants of the surface water are produced from the mining, oil, gas explorations and artisanal/illegal mining. In Peruvian Amazonian, surface water often has high concentration of heavy metals which are leaded downstream of the mines [1].

Furthermore, lakes and rivers receive contamination from animal and human waste. Principally when habitant throw solid, mineral and chemical residues to the environment, rivers, streams or lakes. Therefore, the surface water presents different problems as high turbidity, high concentration of heavy metals, a non-acceptable pH level, high temperature and others. Water contaminated can produce childrens diarrhea and other infections by ingestion and skin disease by dermal absorption. Water contaminated can produce also adverse effect in the plants and animal life [2].

Peruvian Amazon has a large degree of biodiversity, hence a water contaminated put the species in danger of extinction and the risk of damage the Amazonian biodiversity. Aquatic organisms are the principal species in danger with polluting water. Bioaccumulation is used to describe the food chain starting with aquatic organism and fish [1]. People are at the top of this food chain, thus a contaminated water and fish represent also a danger for human consumers.

The Instituto de Investigaciones de la Amazonia Peruana (IIAP) has different research programs, one principal program is the research in the use and conservation of water and their resources (AQUAREC) whose principal objectives are the water monitoring and the sensible use of biodiversity resources [3]. Safeguarding water quality is very important to prevent potential issues occasioned by the contamination. A real time water quality monitoring system allows to prevent and mitigate effect in the shortest term.

In last years, internet of things has achieved great growth for different applications in agriculture, smart cities, health and others. This paper presents the development and implementation of a real time water quality monitoring system design and adapted to Peruvian Amazonian and used in pisciculture.

II. THEORETICAL INFORMATION

SENASA has defined the parameters to controller the water quality to human use and other applications, these parameters are the pH, conductivity, temperature, dissolved oxygen, nitrites, ammonium toxic, alkalinity, hardness etc. However, SENASA also has defined 4 parameters for a fast test of water quality which are the temperature, conductivity, dissolved oxygen and hydrogen ion concentration (pH) [4].

Water temperature and pH have an influence in the behavior and growth of the fish hatcheries. Others parameters are the hardness and dissolved oxygen. Monitoring these 4 parameters allows an optimal control of the water quality for pisciculture in special for the little fish fry. Table I presents the optimal values for different parameters to water quality monitoring. The most common parameters used to care for water quality are:

- **Temperature.** Amazonian fish species need an optimal temperature range to perform their metabolic and physiological processes properly which the favorable nutritional response is between 25°C and 30°C for temperatures outside of this range the fish present fatigue, stress and in some case they die.
- **Dissolved Oxygen.** This value represents the level of free oxygen in water. If this level is too high or too low can damage aquatic life because fish and plants use the oxygen for respiration. The degree of saturation is inversely proportional to the altitude directly proportional to the temperature and pH [5].

TABLE I
PHYSICAL PARAMETERS OF WATER FOR TILAPIA FISH GROWING [6]

Parameter	Optimal range
PH	6.5 - 9
Temperature	24°C - 28°C - 32°C
Conductivity (CaCO ₃)	50 - 350
Oxygen	4.5 mg/L. - 10 mg/L
Ammonium Toxic (NH ₃)	0.01 - 0.1 mg/L
Nitrites (NO ₂ -N)	< 20 mg/L
Alkalinity	0.1 - 0.2 mg/L

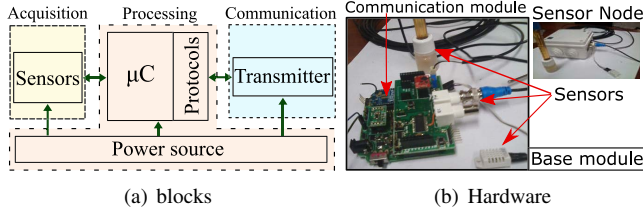


Fig. 1. Sensor node implementation

- **pH.** corresponds to water hydrogen ions concentration. A pH value between 6.5 and 9 is ideal for most species of Amazonian fish. Under 6.5, the fish do not develop completely, that is due principally to disproportionate content of salts in their organism. Values below 4 and above 11 are dangerous for aquatic life.
- **Conductivity.** It is the concentration of mineral compounds in the water, this is proportional to the concentration of metal salts.

III. HARDWARE DESCRIPTION

Sensor node is composed by 3 blocks (Fig. 1(a)): Communication, Processing and Sensor blocks [7] [8].

Three electronic boards were developed; a serial bus communication is implemented in the three boards. This bus allows to send control tasks and data information between each board. Fig. 2 shows the flowchart sequence between the base module, the sensor module and Communication module. For example; the base module starts the process sending a control tasks to activate the sensor module and enable the request data of humidity sensors, temperature and pH. Then the base module activates the GPRS communication card and sends the data to be transmitted to Internet using the cellular network.

A. Base module

This block has a role of controlling and coordinate the actions between the components of the sensor node and performs the synchronization in the communication. This modulo has a microcontroller PIC24F family of 16-bit XLP (Extreme Low Power) which is designed for applications of limited power [9]. Fig. 1(b) shows the Sensor Node designed.

Photovoltaic cells are used to provide energy [10]. Power source uses a BQ24210 circuit to control the limit of voltage and current of photovoltaic cells in the range of 3.7 V and 2500 mA respectively.

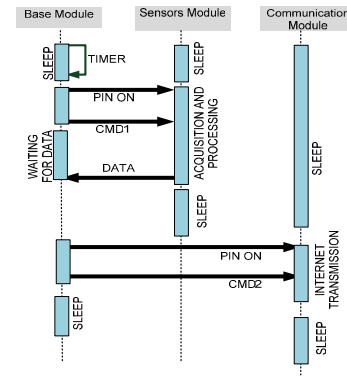


Fig. 2. Illustration of flowchart sequence between the base module, the sensor module and Communication module

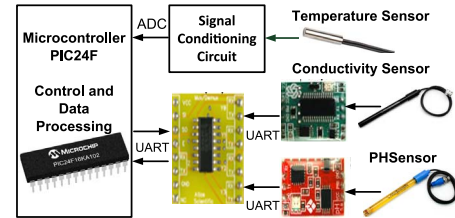


Fig. 3. Diagram of connection of the sensors with the microcontroller

B. Sensor Module

In section below, SENASA has defined 4 parameters, for this sensor nodes 3 parameters are considered: pH, Conductivity and temperature sensor. The role of this module is to acquire the signal of each sensor, and also send this information to the microcontroller of the base module. Fig. 3 shows the connection of sensors and the base module.

- **Temperature sensor.** It is a thermistor with negative coefficient of temperature (NTC). The Steinhart-Hart equation is implemented in the microcontroller to obtain the temperature value from the samples acquired [11].
- **Conductivity.** An Atlas Scientific Conductivity sensor is used, this device has a digital interface based in UART interface.
- **pH.** a glass electrode Atlas Scientific pH sensor is used. Which has a selective ion membrane permeable to hydrogen. The electronic interfaces of the electrodes communicate with the microcontroller using the serial port for sending commands only when the sensor node wakes up from its low power mode.

To control the consumption, the sensors are disconnected from the power supply when the node is in mode sleep, then these sensors are activated when the node wakes up. A Mosfet FDN360 transistor is used to control power supply. Fig. 4 shows the power supply of sensor node.

C. Communication Module

For data transmission 2 option are used: IEEE 802.15.4 standard (ZigBee) and mobile data communication (GPRS)

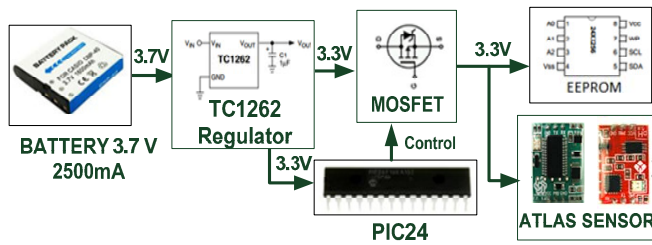


Fig. 4. Basic scheme of energy control using the Mosfet FDN360

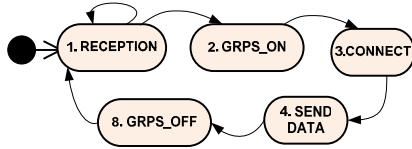


Fig. 5. State Diagram for sending data to Internet

General packet Radio Service). A Digi Xbee module is used (based in IEEE 802.15.4), this has a range up to 100 meters for low-power devices and connection speeds of 250 kbps. It also operates in the ISM frequency band of 2.4 GHz. The activation time of the Xbee device does not require more than 20 milliseconds to start working when leaving the sleeping state, which allows to extend the operating time of a sensor node energized by batteries. The module is used in transparent transmission mode allowing the use of the serial port for sending information from the microcontroller.

This module is in sleep mode until you need to send data to the Internet. The GPRS SIM808 card from the IteadStudio Company allows the sending of data to the Internet for which the firmware and hardware are designed to control its energy consumption and operates on frequencies 850/900/1800/ 1900 MHz. The control of the GPRS module is done by a microcontroller PIC24, which starts its tasks when leaving the "Sleep" mode. The Base module data is received and the GPRS module is connected to the server on the Internet to send the information. Finally the GPRS module is turned off and the microcontroller goes into low power mode. Fig. 5 shows the state diagram of this process.

For communication with the web application, the HTTP POST communication protocol is used, which allows the sending of approximately 60 bytes of information to a web service. Data sent includes information on water temperature, PH, conductivity, battery voltage and solar cell voltage.

IV. WEB MONITORING SYSTEM

A Web application is developed to receive the information from the sensors nodes. Each sensor node uses the HTTP POST protocol to send information to web service. The web service allows to connect the sensor node and the monitoring interface (page web), this application is developed thanks to the open source Node-RED programming tool. The operation mode is based on the communication hardware devices on the Internet of things. This application is designed to receive

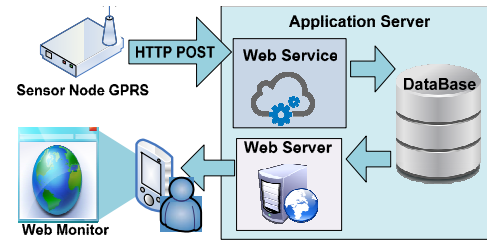


Fig. 6. Communication of the sensor node with the Web application

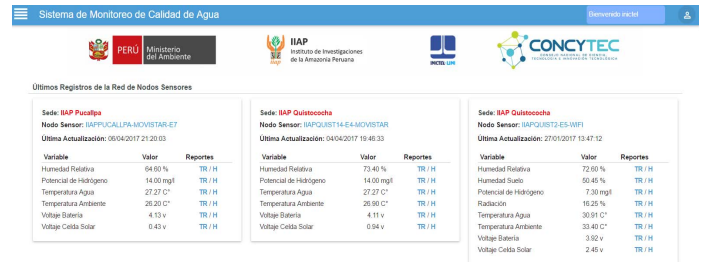


Fig. 7. Web monitoring application using HTTP POST requests

data transparently without knowing if the source is a physical computer or computer system, which makes it very flexible to future updates. Fig. 6 shows the way of the data information from sensor node to web page.

The Web application allows us to access the information of the sensor nodes that will be deployed in three Amazonian regions where the water quality of the breeding ponds will be monitored. Fig. 7 shows the web page to monitoring the six sensor nodes installed in the Peruvian amazon.

Six sensor nodes are installed in different regions of the Peruvian amazon (Fig. 8). To installed the sensor node, an evaluation of the installations and communications of the zone was made in function of the cellular network coverage (for GPRS technology) and the building construction in the zone with internet connection (for ZigBee standard). Three cities of IIAP installation have be chosen:

- **Puerto Maldonado.** Two sensors nodes were installed using Zigbee and GPRS communication technology.
- **Pucallpa.** Two sensor nodes were installed using GPRS communication technology.
- **Iquitos.** Two sensor nodes were installed at IIAP installation of Quistococha using GPRS and ZigBee communication technology.

Fig. 9 shows the data acquired by the sensor node with a transmission period of 6 minutes for 24 hours. Fig. 10 shows the data acquired by the sensor node in the city of Iquitos for 7 days with a transmission period of 30 minutes. During day and night, some changes are observed in the temperature, humidity and pH data. A test of energy consumption for different transmissions periods is presented in Table II.

Fig. 11 shows the voltage variation in the battery and solar cell during day and night. When the battery reaches its maximum load (on the second day) its voltage level is almost 4.2V while the voltage of the solar cell reaches almost 6V. This



Fig. 8. Location of sensor nodes in Peruvian amazon

TABLE II
WIRELESS SENSOR NODE TRANSMISSION PERIOD

Battery [mAh]	Tx period [s]	I _{run} [mA]	t _{run} [s]	I _{sleep} [mA]	t _{sleep} [s]	Duration [days]
2200	20	40	6	0.4	14	5.59
2200	60	40	6	0.4	54	15.76
2200	300	40	6	0.4	294	57.67
2200	600	40	6	0.4	594	86.36

is because the load circuit disconnects solar cell and causes reach its maximum voltage in open circuit avoiding overload.

V. CONCLUSION

Six Sensor nodes are deployed in three cities of the Peruvian Amazon (Iquitos, Puerto Maldonado and Pucallpa). These locations were coordinate with researchers of the IIAP. The monitoring system presented allows to study the water quality parameters applied to pisciculture; moreover, this system could also be applied for environmental monitoring of the flora and fauna. For the implementation of the sensor node, we have considered energy consumption and communication requirements. Laboratory tests of sensor node show a highest current consumption in the wireless transmission modules in comparison with the others modules. Data acquired is transmitted by GPRS and Zigbee protocols. HTTP POST protocols allows a remote access to data acquired by sensor

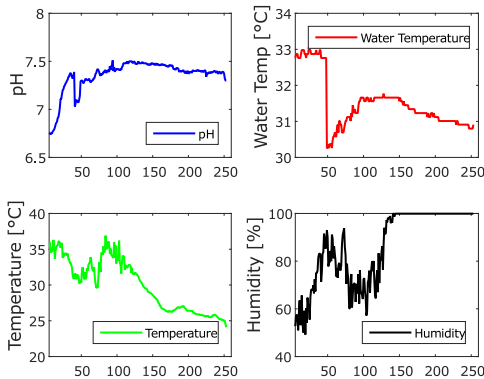


Fig. 9. Sensor node located in Iquitos

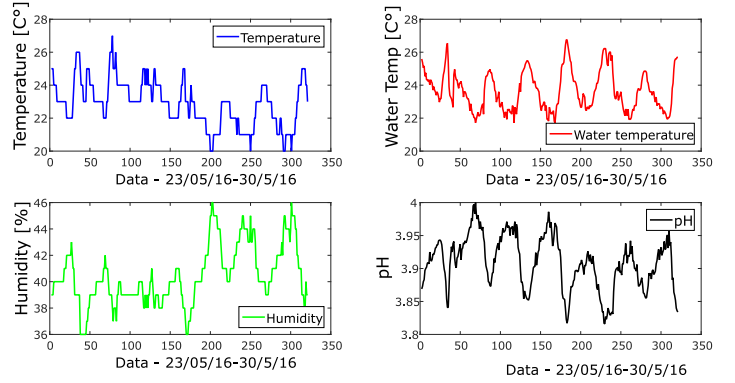


Fig. 10. Sensor node located in Pucallpa

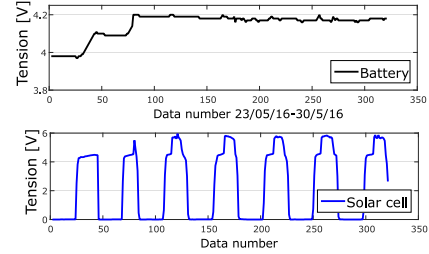


Fig. 11. Voltage in Sensor node located in Pucallpa

node. All data are used by the researchers of IIAP to control the pollution of water in pisciculture.

Future work should be focused in the study of the source coding and inference algorithms to decrease the energy consumption in the sensor node.

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